

T.I.P MART

Mobile App

PROJECT TYPE Mobile App	PLATFORM Native Android application
AUDIENCE T.I.P. campus buyers and sellers	PRIMARY FOCUS Student-to-student marketplace

Overview

A modern Android marketplace application designed specifically for students of the Technological Institute of the Philippines (T.I.P). Built with Kotlin and Jetpack Compose, featuring secure authentication, smart pickup system, and comprehensive e-commerce functionality.

Core Features

- Secure Authentication with Firebase
- Shopping Cart & Wishlist functionality
- Smart Campus-based Pickup System
- In-app Messaging between users
- Multiple Payment Methods (GCash, Maya, Bank Transfer)
- Real-time Order Tracking & Notifications
- Seller Dashboard with Analytics
- Rating & Review System

Experience and Design

- Campus-focused discovery keeps listings relevant to the student community.
- Cart, wishlist, checkout, and order tracking form a complete buyer journey.
- Seller tools support listing creation, inventory monitoring, and order fulfillment.
- Messaging, ratings, reviews, and notifications create trust between buyers and sellers.

Architecture and Data

- Kotlin and Jetpack Compose provide the native Android presentation layer.
- MVVM separates composable screens, view models, repositories, and Firebase services.
- Cloud Firestore stores listings, orders, messages, reviews, and user records in real time.
- Firebase Storage handles listing media, while Cloud Messaging delivers notifications.

Security and Privacy

- Firebase Authentication protects account access and user identity.
- Firestore security rules scope marketplace records to authorized operations.
- Storage rules restrict uploaded listing media to authenticated users.
- Client-side validation and typed models reduce malformed listing and checkout data.

Technology Stack

Kotlin	Application language
Jetpack Compose	Application framework and UI
Firebase	Authentication, database, or cloud data
Material Design 3	Interface styling and design system
MVVM	State, data flow, and architecture
Coroutines	State, data flow, and architecture

Performance and Reliability

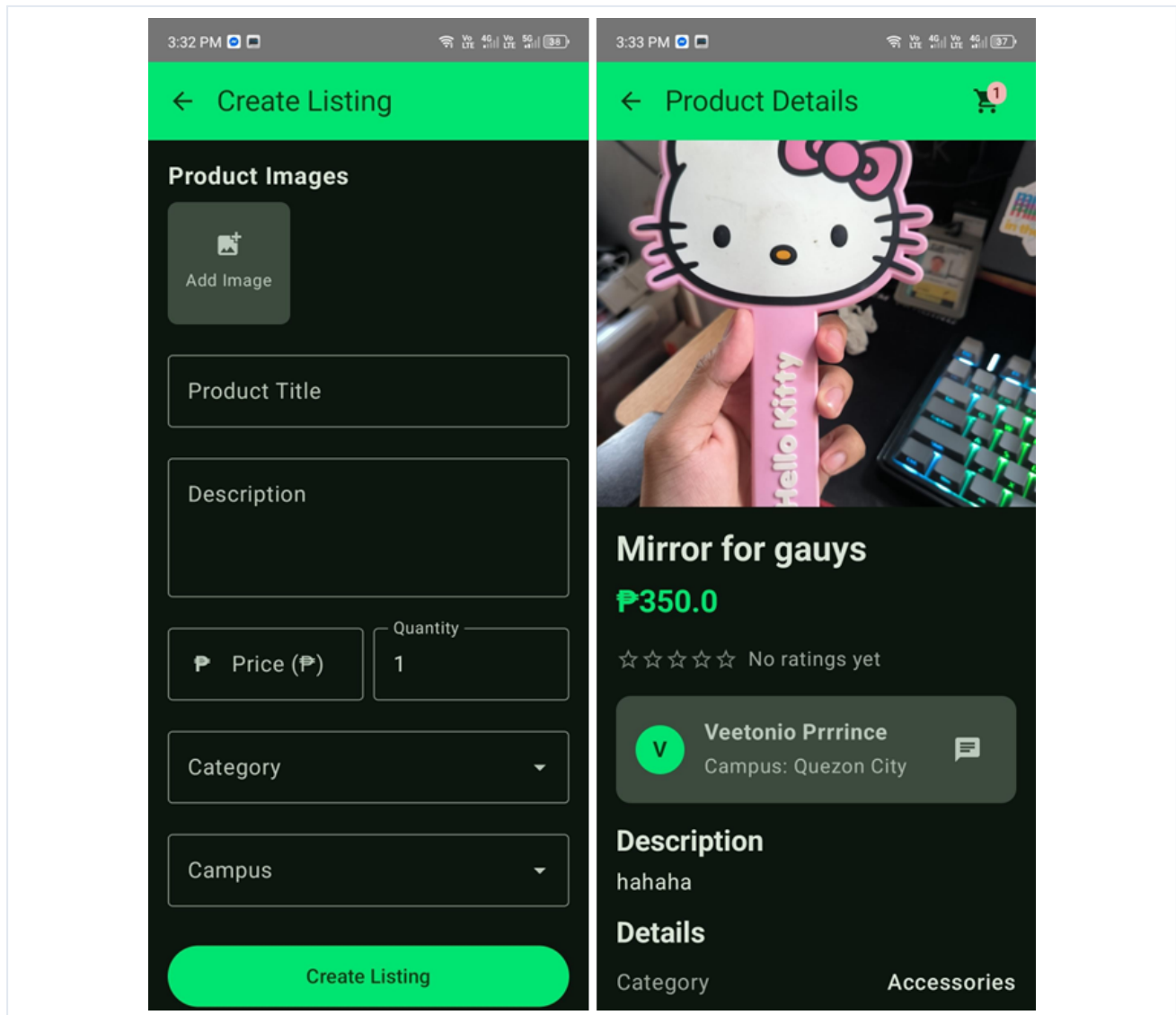
- Coroutines keep network and database work off the main thread.
- Reactive state updates refresh listings and orders without manual polling.
- Repository-level caching limits repeated Firebase reads.
- Compose list virtualization keeps catalog and order screens responsive.

Platform Requirements

- Android device with Android 5.0 or later.
- Internet connection for authentication, listings, messages, and order updates.
- Storage or camera access when adding product images.
- A valid T.I.P. student account for the intended marketplace experience.

PAGE OVERVIEW

Create Listing and Product Details

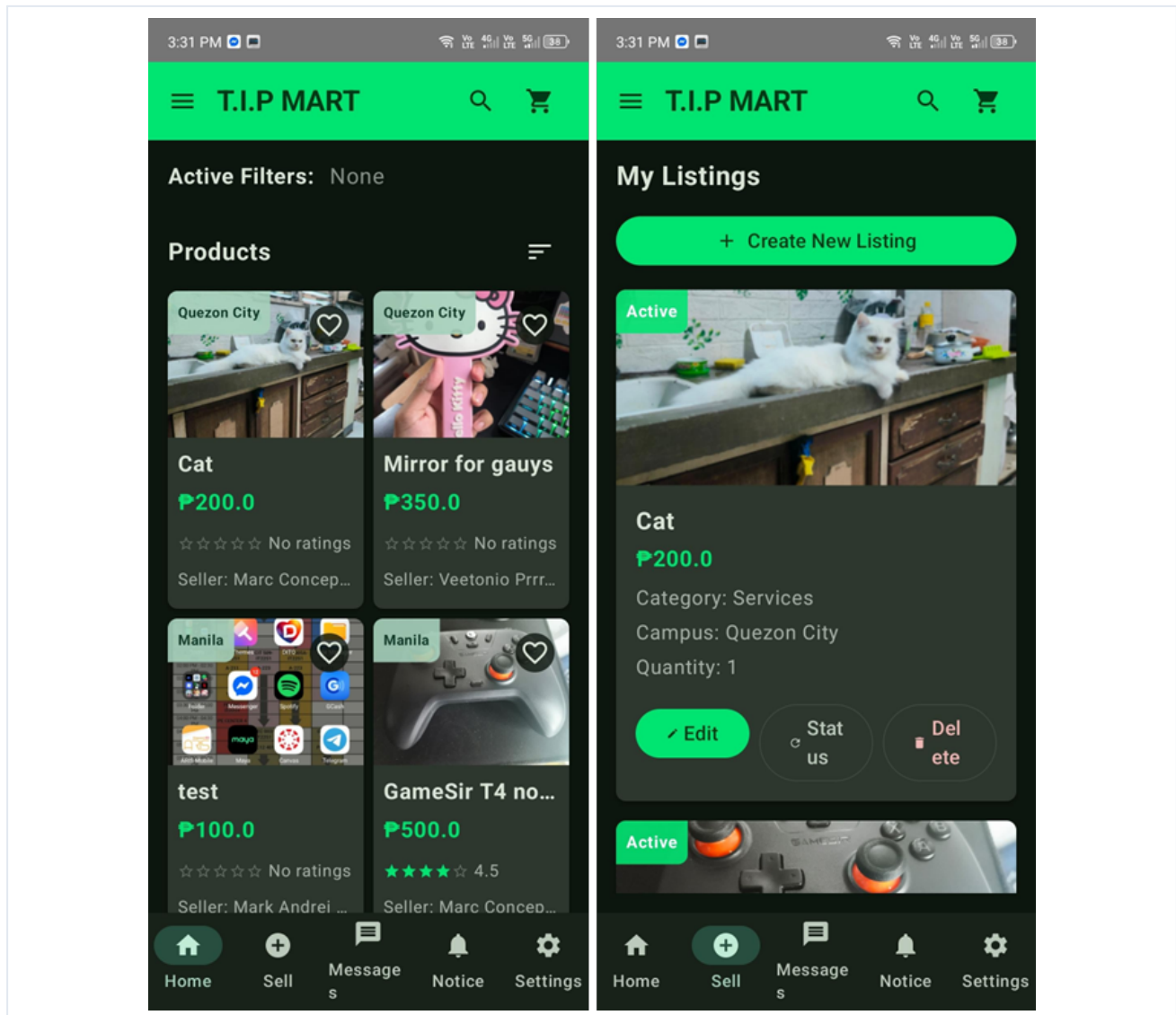


Create Listing and Product Details

Interface view from T.I.P MART.

PAGE OVERVIEW

Home & My Listing

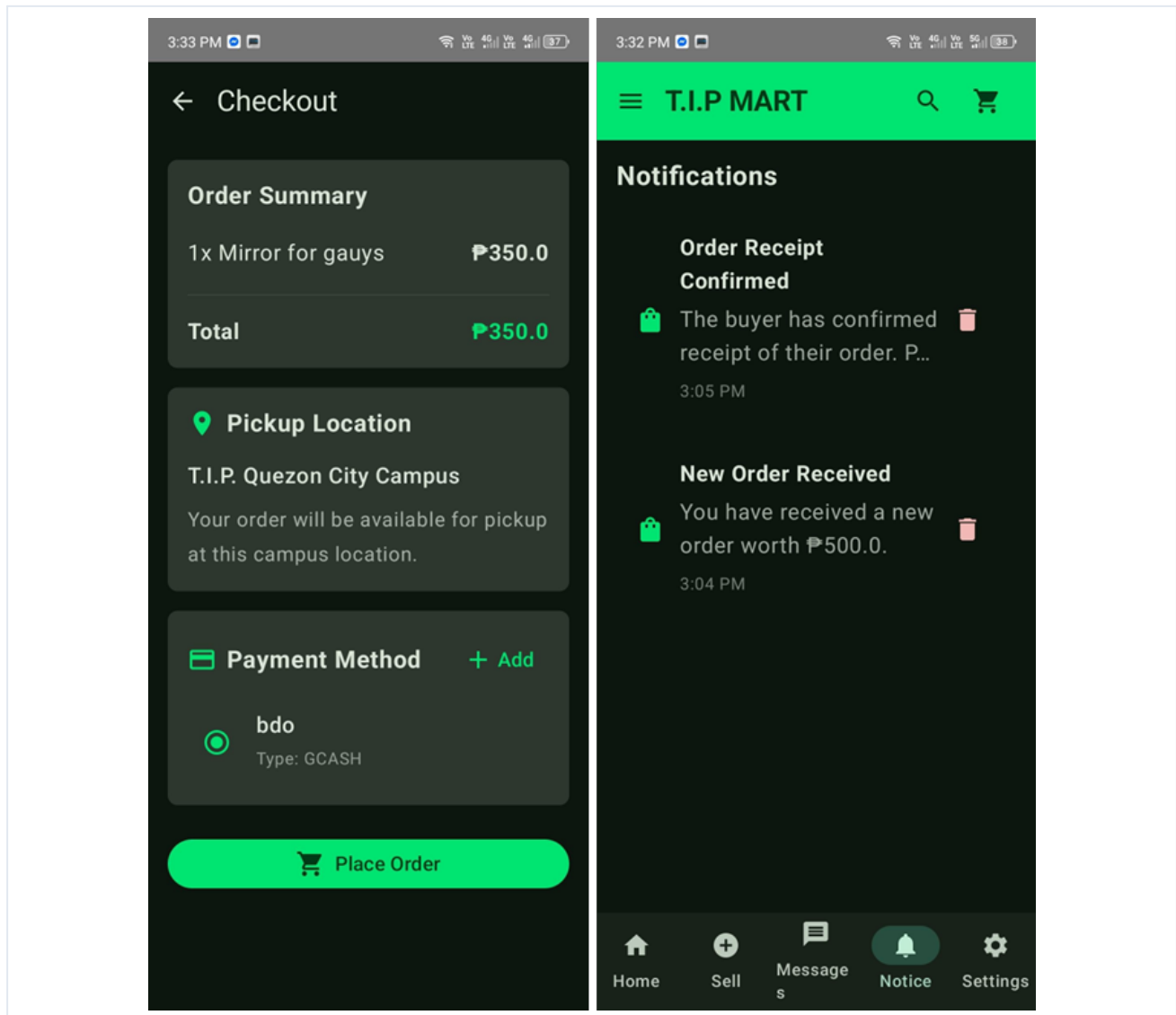


Home & My Listing

Interface view from T.I.P MART.

PAGE OVERVIEW

Checkout & Notifications

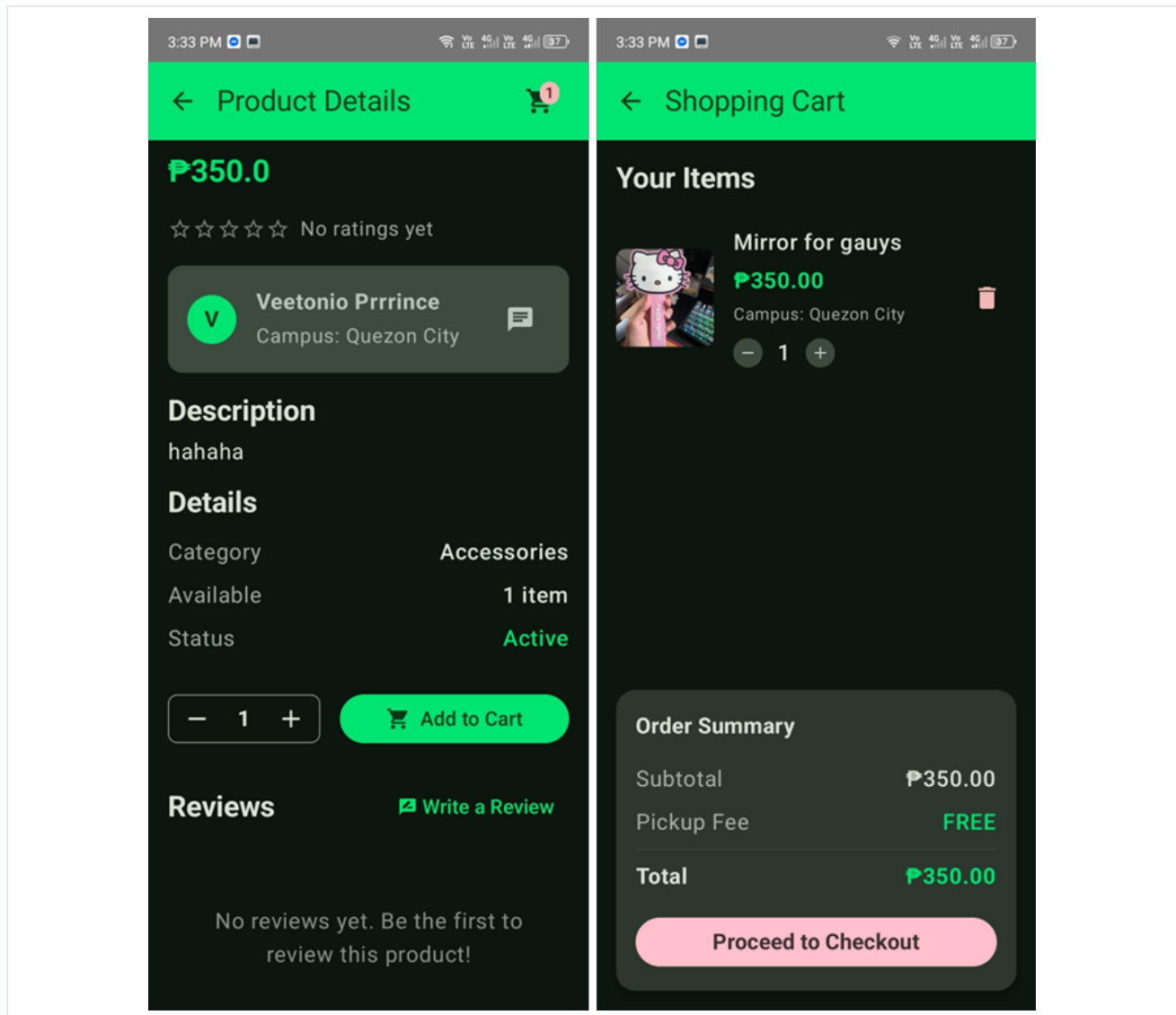


Checkout & Notifications

Interface view from T.I.P MART.

PAGE OVERVIEW

Product Details & Shopping Cart

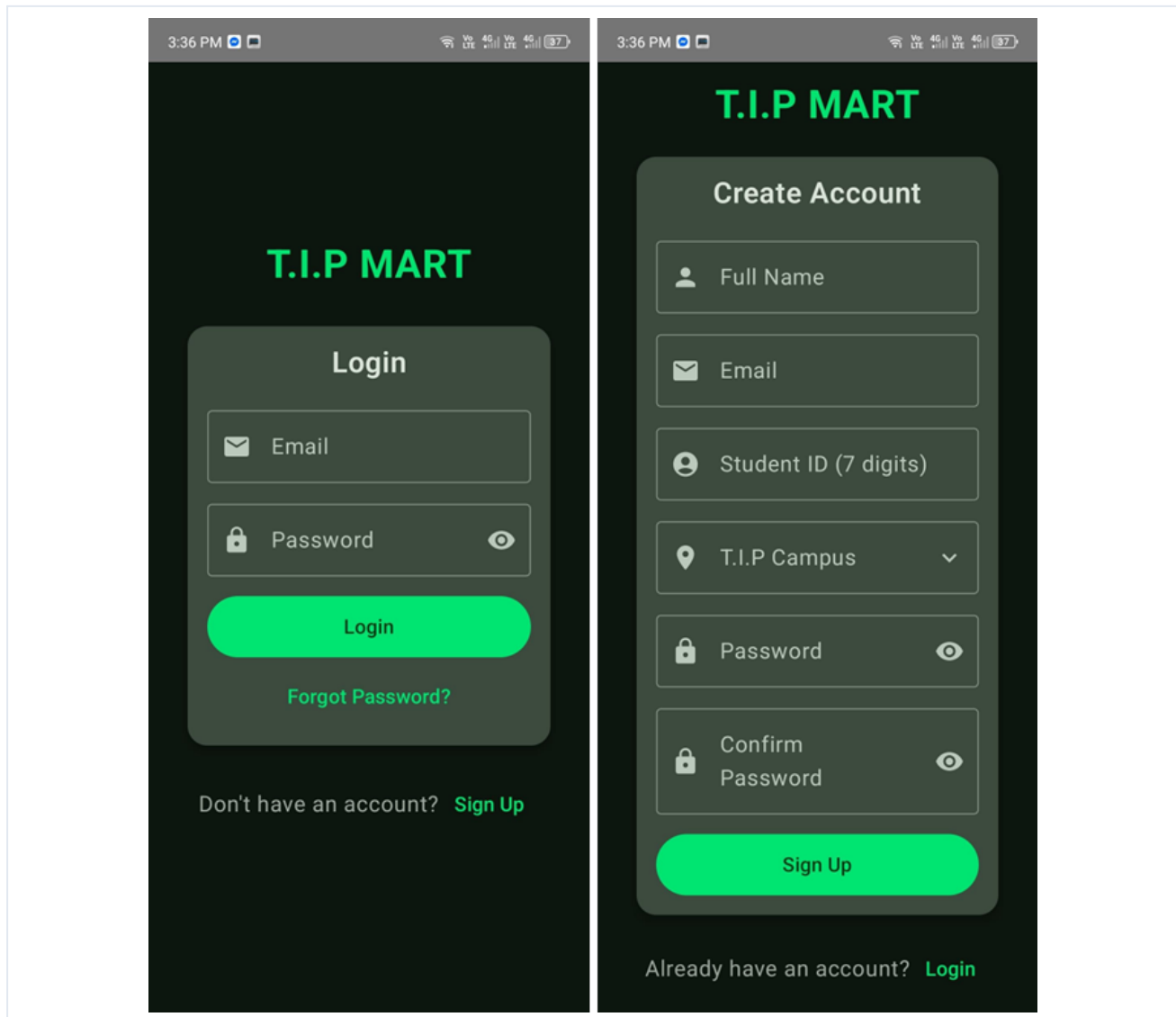


Product Details & Shopping Cart

Interface view from T.I.P MART.

PAGE OVERVIEW

Login and Sign up

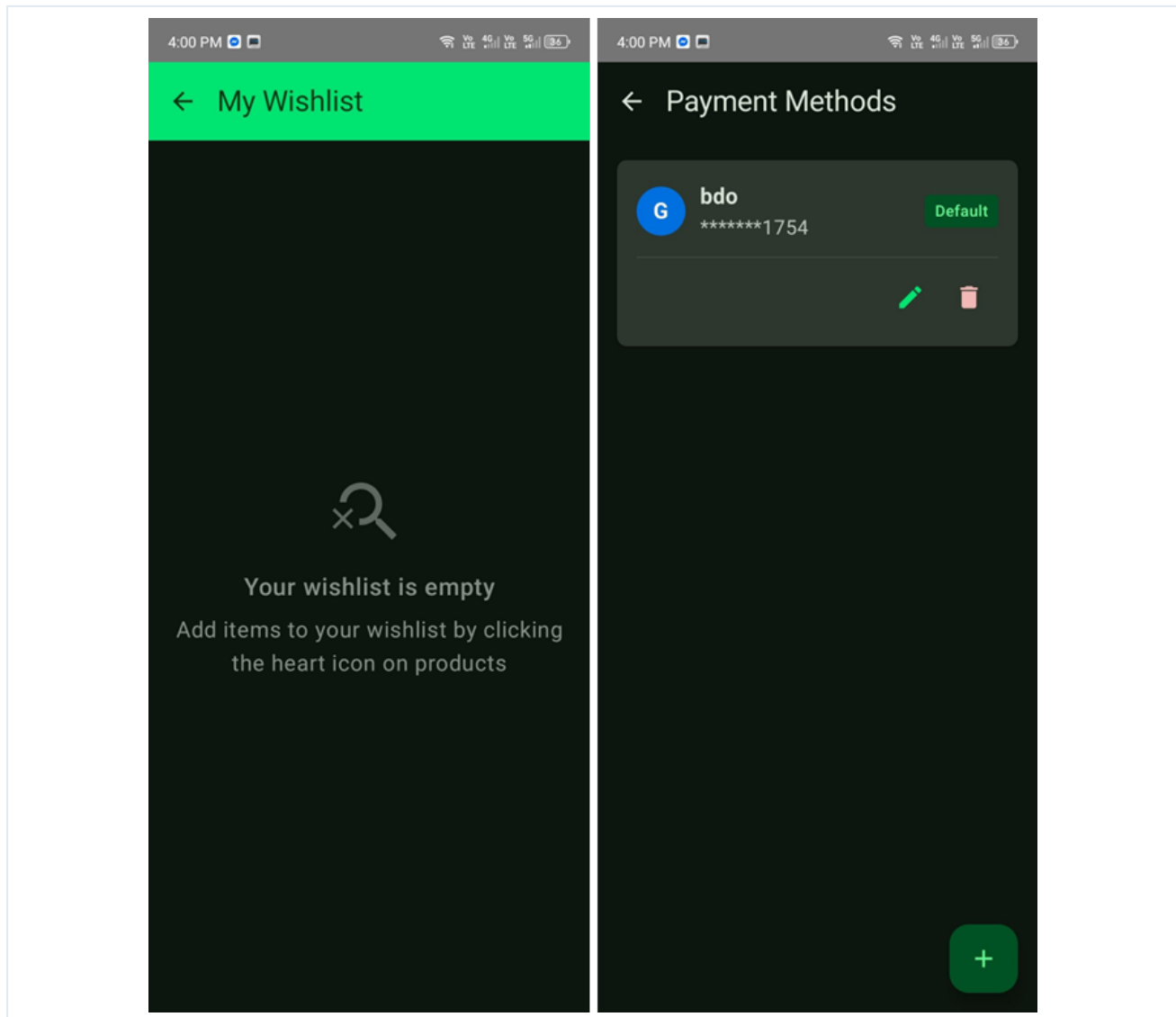


Login and Sign up

Interface view from T.I.P MART.

PAGE OVERVIEW

Wishlist & Payment Methods

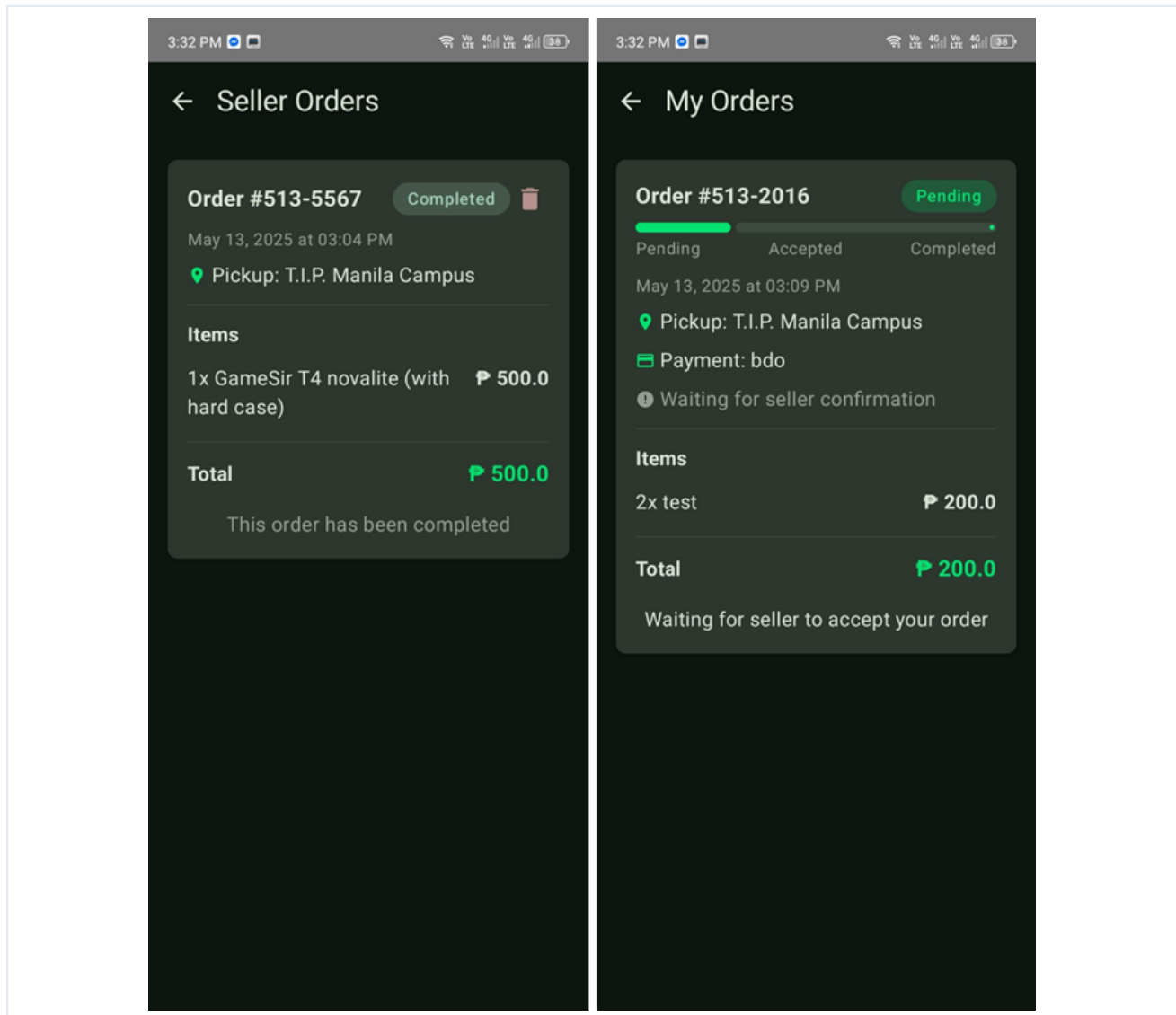


Wishlist & Payment Methods

Interface view from T.I.P MART.

PAGE OVERVIEW

Seller & My Orders

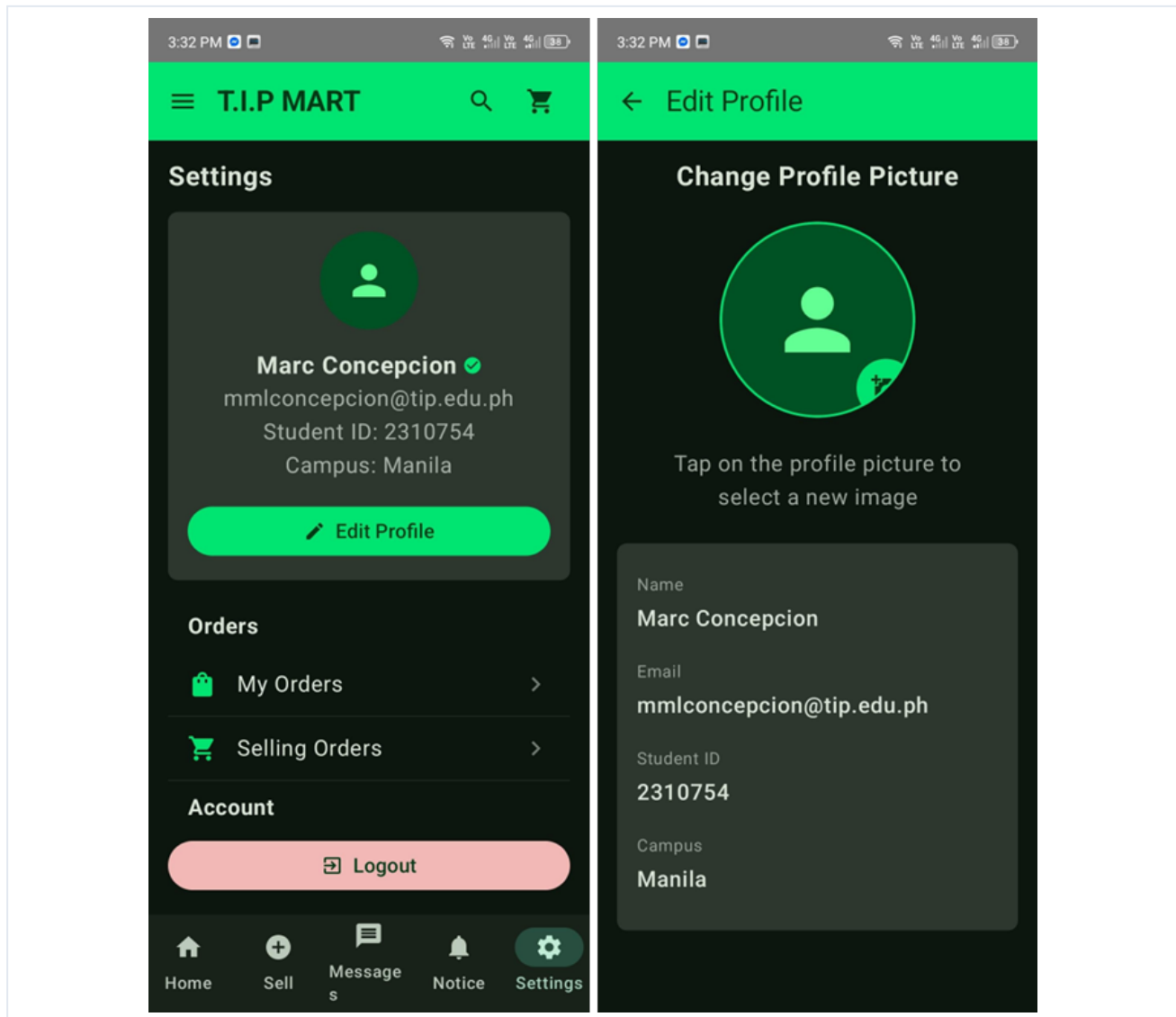


Seller & My Orders

Interface view from T.I.P MART.

PAGE OVERVIEW

Settings and Profile



Settings and Profile

Interface view from T.I.P MART.